

PURELY COSMETIC SURGERIES VIA A PU(2)-MONOPOLE COBORDISM: A DIGEST OF THE ARGUMENT

Setting. $Y_r = S_r^3(K)$. By Hanselman and Daemi–Lidman–Miller Eismeier (Cor. 1.4), the only open case is $g(K) = 2$ with an orientation-preserving diffeomorphism $\phi: Y_{-2} \rightarrow Y_2$. Everything below derives a contradiction from ϕ ; no peripheral condition on ϕ is used.

Strategy. Use ϕ to build a closed 4-manifold X with an n -cube of metrics and a signed sum over 2^n determinant lifts such that (a) the cut-down zero-dimensional instanton count is an integer $\Omega \neq 0$, and (b) the coupled Dirac index $n_D = \Theta - \kappa$ is positive. Run the PU(2)-monopole cobordism (Pidstrigach–Tyurin, Okonek–Teleman, Feehan–Leness): after the S^1 -quotient and $\eta = n_D - 1$ phase cuts (sections of the weight-two line) the free monopole space is 1-dimensional. Its ends are instanton links ($2^n\Omega$), Seiberg–Witten-type reducibles, and face ends. Lattice and curvature estimates remove *all* reducible fixed points and all face ends except lens walls. The lens walls cancel in pairs with the same signs that make Ω integral. Hence $2^n\Omega = 0$. The analysis never glues at a Seiberg–Witten point or at a bubbling level.

1. Ordinary maps (paper, Secs. 2–3)

Use the irreducible determinant-one complexes. $Y_{\pm 1}, Y_{\pm 2}, S^3$ have only central reducibles. On Y_0 , $w \cdot A$ is odd (A the capped Seifert surface). The unperturbed complex gives $C(S^3) = 0$.

- $f_+: Y_0 \rightarrow Y_1 \rightarrow Y_2$ (meridional 2-handles) and its mirror $f_-: Y_{-2} \rightarrow Y_0$ are $\mathbb{F}_2$ -isomorphisms by exact-triangle pentagons (Culler–Daemi–Xie, DLME); the third term vanishes because $C(S^3) = 0$. The outer cap is a sphere chain bounded by $S^1 \times S^2$. Replacing it by $S^1 \times D^3$ gives continuation, and the harmonic path of boundary flats has odd degree. $g_+: Y_2 \rightarrow S^3 \rightarrow Y_0$ and g_- are one-parameter interval counts, summed mod 2 over two lifts differing by PD of the (-2) -sphere; the $\mathbb{RP}^3$ -wall terms agree and cancel mod 2. The compressed pentagon gives $f_+g_+ \simeq \text{id}$.
- *Negative interval B (first new ingredient).* $W' = (-C_2) \cup_{S^3} C_{-2}: Y_2 \rightarrow Y_{-2}$ has $b^+ = 0$ and $H_2 = \langle F_l, F_r \rangle$ with $F^2 = -2$. The sphere is $S = F_l - F_r$, $S^2 = -4$. The metric interval runs from the S^3 -split J to splitting off $N = \nu S$, $\partial N = L(4, 1)$. The degree-two insertion $x(S)$ requires the normalized holonomy $= -1$ on a one-parameter sweep of loops across S . The count has $i + 1 + 1 - 3 = 0$ and sums over lifts $c_e = e \text{PD}(S)$. At the lens wall only the minimal cap survives: trace-zero flat, $\kappa = \frac{1}{4}$, harmonic reducible. Its eigenline has degree one, so the sweep crosses -1 with signed count ± 1 . A local determinant-line comparison (tensor by the line $-2x$ on N , then a boundary Weyl conjugation) gives weights $\varepsilon_1 = -s\varepsilon_0$ that cancel these terms *integrally*. Hence B is an integral chain map. Mod 2, $B \simeq g_-g_+$: in the 3-dimensional associahedron for $2 \rightarrow \infty \rightarrow 0 \rightarrow \infty \rightarrow -2$, the face νU ($U^2 = 0$) becomes B in two steps. First replace νU by $S^1 \times D^3$; the w_2 root mismatch makes the matching condition $\text{Hol}_\beta = -1$. Then contract β through two discs whose union is $\pm S$.
- Bridge $T = f_- \phi^{-1} f_+: I(Y_0) \rightarrow I(Y_0)$ is a mod-2 unit, hence a $\mathbb{Q}$ -isomorphism.

2. Caps with a nonzero pairing on Y_2 (paper, Sec. 4)

Take a symplectic closure $X_0 \supset Y_0$ (Gabai, Eliashberg–Thurston, Eliashberg caps, as in Kronheimer–Mrowka). Arrange simply connected halves, tight genus-2 surfaces in each half, and A primitive. $D_{X_0}^w(x_0^m h^a) \neq 0$ with $h \cdot A = 1$; the paper reproves this. The extremal joint eigenspace of $(\mu(\Sigma), \mu(\text{pt}))$ on $I(\Sigma_g \times S^1)$ is a line (Muñoz). The Lefschetz word of the blown-up X_0 is extracted by Hurwitz moves as a prefix of model words from $E(g+1)$, which have nonzero extremal coefficient. Then apply the KM structure theorem and the blow-up formula. Insert T^k at Y_0 . Surface insertions crossing T are tracked by a nilpotent deformation over $\mathbb{Q}[t_i]/(t_i^2)$, and the deformed T_L is still a quasi-isomorphism. Cayley–Hamilton gives $k \geq 1$ with nonzero all-label coefficient. Cut at an internal Y_2 seam, where H_2 splits over $\mathbb{Q}$, and allocate the probes. This gives caps $C_\pm$ with $b^+ > 0$, containing A with $w \cdot A$ odd, so generic periods kill reducibles. Each cap gets one extra blow-up E_{exc} . The result is an integral cycle x and cocycle y on $I(Y_2)$ with $yx = q \neq 0$.

3. The stack: $\Omega \neq 0$ and $n_D > 0$ (paper, Sec. 5)

$X = C_- \cup (n \text{ copies of } B, \text{ each followed by } \phi) \cup C_+$, except at m places where $H = f_+f_-$ replaces the return. These positive bridges are spaced exactly four intervals apart.

- *Integrality.* Each step is a unit of $L/2^N L$, where $L = I(Y_2; \mathbb{Z})/\text{Tor}$ and $q \neq 0$ (2^N). Take all repetition counts divisible by $\exp \text{GL}_r(\mathbb{Z}/2^N)$ and by 8. Then the composite is the identity mod 2^N , so $\Omega \equiv \pm q \neq 0$. Stretching the finite Y -seams identifies Ω with the composite: a central contact costs 3, and a macro BHB recovers at most 2 per central outer limit.

- *Lattices.* Cut at the S^3 's. A negative cell is unimodular, $\langle a, b \rangle$ with $a^2 = b^2 = -1$. A positive cell has $H_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \oplus (-1)^4$, $b^+ = 1$, $\sigma = -4$, and contains the plumbing $X_1 = -S_j$, U , $X_2 = S_{j+1}$ with squares $(-4, 0, -4)$.
- *Indices.* Fix a characteristic l and put $\Lambda = l + c$. A bit flip adds $\text{PD}(S_i)$ to both c and Λ , preserving Λ^2 and κ . $\Theta = (\Lambda^2 - \sigma)/4$ telescopes, gaining $+1$ per positive cell. With $D_I = 8\kappa - 3(1 + b^+) = n + z$ and $b^+ = m + b_0$, $n_D = \Theta - \kappa = \frac{5m-n}{8} + \text{const}$, which is about $m/8$ for $n \approx 4m$.

Where ϕ enters. $n_D > 0$ needs $m/n > 1/5$. The exclusions of part 5 below need spacing ≥ 4 , i.e. $m/n \leq 1/4$. Without ϕ every B must be followed by H , since the reversed interval $Y_{-2} \rightarrow Y_2$ has $b^+ = 2$. That forces $m \approx n$.

4. Metrics and the Abelian clamp (paper, Secs. 6–7)

Only the S^3 cuts J_i and the lens cuts ∂N_i become infinite; both have psc. All Y -seams stay finite, so no monopole Floer theory on Y_j is needed. In each macro the two adjacent interval parameters drive a toric Kähler metric (Guillemin) on the $(-4, 0, -4)$ plumbing. Its moment polygon is $\{4|y| \leq x \leq 1, -a \leq y \leq b\}$, its class is $H = U + aX_1 + bX_2$, and $0 \leq a, b \leq \frac{1}{4}$. The value $a = \frac{1}{4}$ splits off N_1 (the lens face); $a \rightarrow 0$ approaches J . Abreu's formula plus a Schur product identity give $\text{Scal} \geq 0$. Near $a = b = 0$ a round $S^2 \times S^1$ tube in class U has length $\sim 1/\epsilon$. For reductions (R: $\Phi = 0$ split; S: Φ in one summand) let v be the line difference, $K = \Lambda + v$ and $u = v \cdot U$.

- Small width: the energy is $o(1/\epsilon)$, but $u \neq 0$ costs $\gtrsim u^2/\epsilon$. Hence $u = 0$.
- Positive width: Weitzenböck gives $vH = 0$ (R) or $(vH)(KH) < 0$ (S). H is a convex combination of the vertex classes $H_I = U + \frac{1}{4} \sum_{i \in I} X_i$, and $\Lambda H_I < 0$. So for some $I \neq \emptyset$: $\text{sign}(u) vH_I \leq 0$ (R) or $< |\Lambda H_I|$ (S).
- Caps: no R component meets a cap. S components satisfy v_W^2 , $K_W^2 \leq C_W$, with C_W independent of the stack.

5. Index exclusion (paper, Sec. 8)

The equations are $D_a \Phi = 0$, $\rho(F_a^{0,+}) = (\Phi \Phi^*)_{00}$, with determinant-one gauge and S^1 acting on Φ . Fixed types are A (instanton), R and S; free type is P. Take a limit with k cuts at the J_i . Virtually fill the lens caps with sharp references; the framed lens corrections come from $\mathbb{F}_4$ versus $\mathbb{P}(1, 1, 4)$. Then $q_i = D_{I,i} + p_i - 2s_i - z_i$ satisfies $\sum (q_i + 4) = 4$. On R/S components $\kappa_i = -v_i^2/4 + \ell_i$ with $\ell_i \geq 0$. A test with $v(S) = 0$ on a reducible forces a unit of ℓ ; this uses a smooth zero-winding modification of the holonomy test. Every assigned test pays a negative half of square ≤ -2 or a unit of ℓ . Every positive cell has square ≤ -2 by the clamp, with one exception that is paid by ℓ . With $L \geq 4m - 3$ (spacing four), every interior run of R/S components has $\sum (q_i + 4) \geq -2$. A cap component is never R. An S cap component has $q + 4 \geq 8$ after long delays and large odd $\Lambda(E_{\text{exc}})$, using both cap bounds. So $k \geq 1$ would force $\sum (q_i + 4) \geq 2a + 6b + 2 \geq 6 > 4$, and $k = 0$ forces type A. Hence there are no Seiberg–Witten (S) or R fixed points at all. For a mixed limit, delete the R components and keep their parameters. For $k \geq 1$ there are $N \geq 2$ non-R components and the projected dimension is $\leq 3 - 2N < 0$. The only survivor is one P component plus one split N_i with the minimal trace cap ($\kappa = \frac{1}{4}$) and no particles.

6. Boundary count (paper, Secs. 9–10)

Instanton links: make the Dirac operator surjective; the link is $\mathbb{CP}^{n_D-1}$ and the phase cuts are $\mathcal{O}(2)$ -sections, giving 2^n sign. Lens ends: on N the spinor vanishes and the Dirac operator is invertible (psc, index 0), and the cap glues to the free main solution. The two bit lifts contribute opposite terms with the same ε as in B ; the Dirac line is complex, so it adds no sign. Stokes (rational multisections on the free stratum) gives $0 = \pm 2^n \Omega$, a contradiction. $\square$

Assessment

The architecture is the standard PU(2)-monopole cobordism. Its twist is that the geometry empties the Seiberg–Witten side, so the Feehan–Leness reducible gluing is never needed. Genuinely new: (i) the integral negative-interval map B , whose lens-wall cancellation carries over to the monopole problem; (ii) using ϕ to set the positive-bridge density in $(\frac{1}{5}, \frac{1}{4}]$; (iii) the toric psc families and the period clamp on the $b^+ = 1$ cells. The mod- 2^N repetition and the labelled deformation are routine. The rest of the length is index bookkeeping and analytic verification. Points to check, in my reading:

- transversality and compactness of the “necessary projection” that forgets R components in mixed strata, with multisections;
- the zero-winding holonomy test: it must preserve the class of B and behave correctly under Uhlenbeck limits;
- uniformity of the clamp across the degenerating toric family and genuine faces, and its compatibility with the product-metric computation of Ω ;
- the local orientation comparison that cancels lens ends in both problems.